Draw ER Diagram and DFD Diagram

ER model - Database Modeling

- The E-R model is very useful in **mapping the meanings and interactions of**
- **real-world** enterprises onto a conceptual schema.
- It facilitates database design by allowing specification of an **enterprise schema** that represents the overall logical structure of a database.
- The ER data model employs three basic concepts/components:
  - **Entity Sets**
  - **Relationship Sets**
  - **Attributes**
- The ER model has an associated diagrammatic representation, the **ER diagram**, which can express the overall logical structure of a database graphically.

Entity

- A **real-world thing** either living or non-living that is easily recognizable. It is anything in the enterprise that is to be represented in our database.
- It may be a **physical thing or simply a fact** about the enterprise or an event that happens in the real world like place, person, object, event or a concept.
- **Examples of entities:**
  - **Person:** Employee, Student, Patient
  - **Place:** Store, Building
  - **Object:** Machine, Product, Car
  - **Event:** Sale, Registration, Renewal
  - **Concept:** Account, Course

Entity Sets

- An entity is represented by a set of attributes; i.e., descriptive properties possessed by all members of an entity set.
- An **entity set** is a set of entities of the same type that share the same properties or attributes. It may contain entities with attribute sharing similar values.
  - Example:
    - **Faculty** entity set – set of all faculty in college
    - **Student** entity set – set of all students in college
    - **Course** entity set – set of all courses in college
- Entity sets do not need to be disjoint.
  - For example, it is possible to define the entity set of all people in a college like person.
  - A **person** entity may be an faculty entity, a student entity, both, or neither.

Attributes

- Each entity is described by a set of attributes/properties.
- Attributes are descriptive properties possessed by each member of an entity set.
- All entities in an entity set/type have the same set of attributes.
- **Student** entity is described by
  - Student_ID–ID of the student.
  - Student_Name –name of the student.
  - Student_Class–class in which student study
- Each entity has a **value** for each of its attributes.

Student
Student-ID
Student_Name
Student_class

Student

Student_ID	**Student_Name**	**Student_Class**
S101	Sachin Tendulkar	First Year
S102	Rahul Dravid	Final Year

Key Attributes

- Key is an attribute or a collection of attributes whose value(s) **uniquely identify** an entity in the entity set.
- That is two entities in an entity set are **not allowed** to have exactly the same value for all attributes.
- The **primary key** of an entity set allows us to distinguish among the various entities of the set.
- Example:
  - Roll_Number is the primary key for Student entity set
  - Emp_ID is the primary key for Employee entity set

Relationships

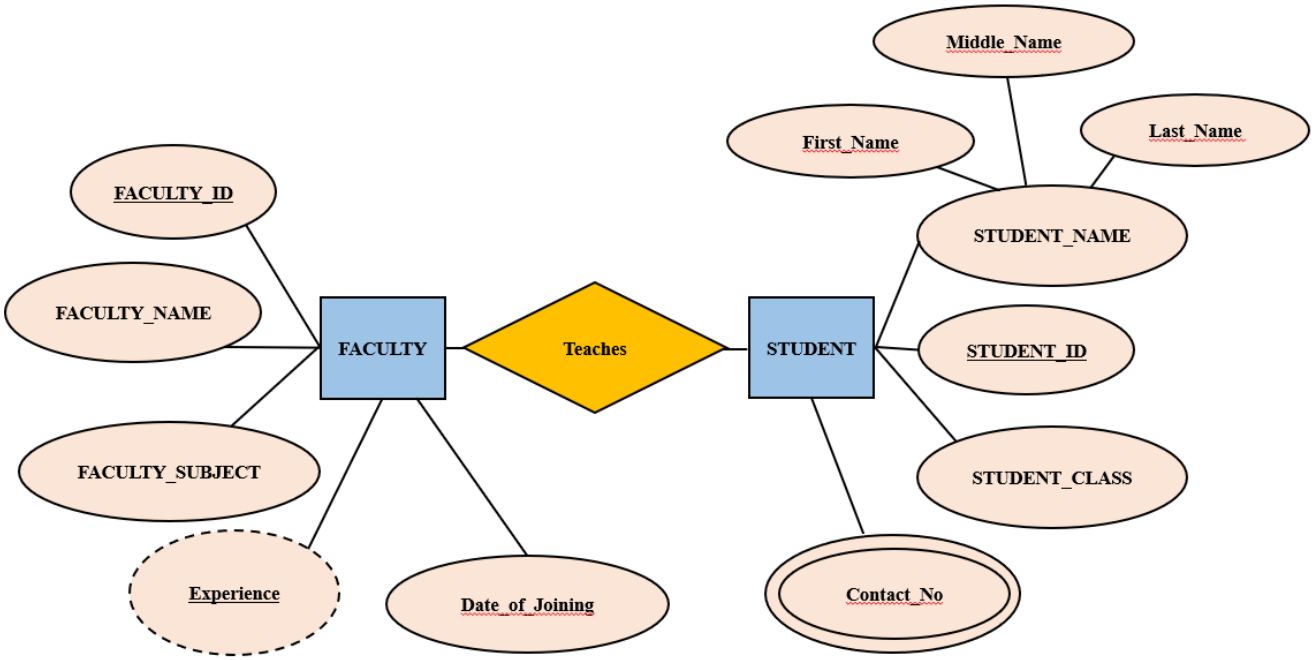
- A **relationship** is an association among several entities. Entities take part in relationships.

  Example:
  - student Rahul Dravid enrolls in Data Science course

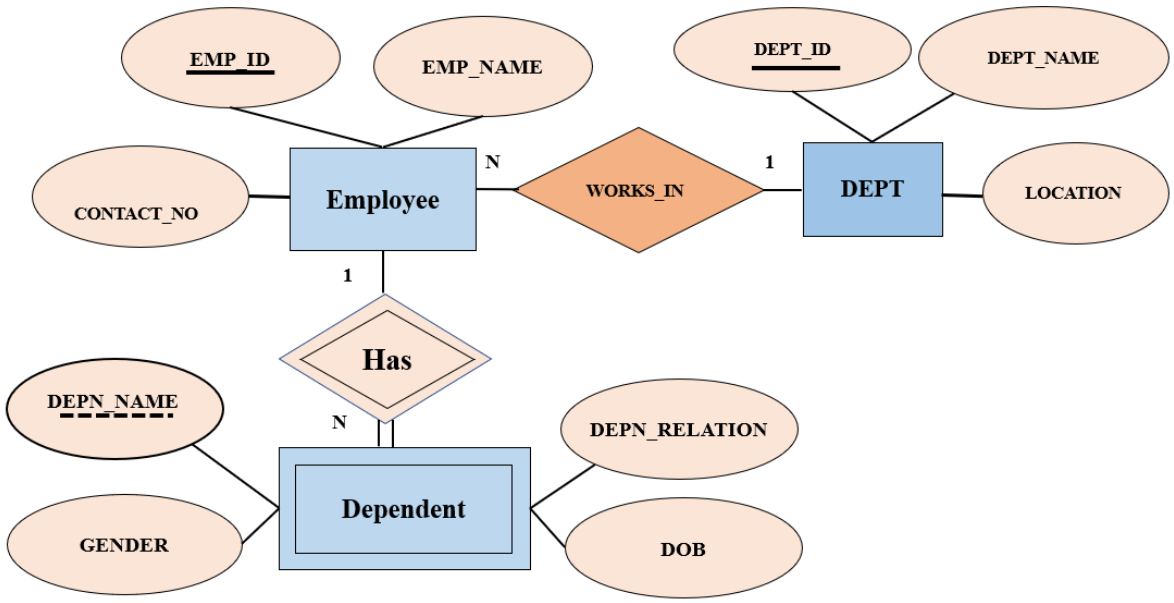
  Relationship **enroll** has Student and Course as the **participating entity sets.**

Example-Entity Relationship (ER) model

College Database



Employee Database



Data flow diagram

A **data flow diagram** (**DFD**) is a **graphical representation of the "flow" of data** through an information system, modelling its process aspects.

A DFD is often used as a **preliminary step to create an overview of the system without going into great detail**, which can later be elaborated.

A DFD shows **what kind of information** will be input to and output from the system, **how the data will advance** through the system, and **where the data will be stored**.

It **does not show information about** process timing or whether processes will operate in sequence or in parallel.

Levels in Data Flow Diagrams (DFD)

DFD can be drawn to represent the system of different levels of abstraction.

Level-0 DFD (context diagram):

It is also known as a **context diagram**. It's designed to be an **abstraction view**, showing the **system as a single process** with its relationship to **external entities**. It represents the entire system as a single circle with input and output data indicated by incoming/outgoing arrows.

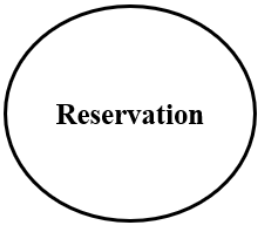
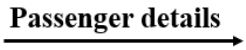
Level-1 DFD:

In level-1 DFD, the context diagram is decomposed into **multiple sub-processes**. In this level, we highlight the main functions of the system and **breakdown** the high-level process of level-0 DFD **into sub-processes**.

Level-2 DFD:

Level-2 DFD goes one step **deeper into parts** of level-1 DFD. It can be used to plan or record the **specific/necessary detail about the system's functioning**.

Graphical notations for Data Flow Diagram

Term	Notation	Explanation
External entity		External entities are represented by a **rectangle** and the name of the external entity is written inside the rectangle.
Process		Processes are represented by **circle** and name of the process is written inside the circle.
Data store		A left-right **open rectangle** is denoted as data store; name of the data store is written inside the shape
Data flow		Data flow is represented by a **directed arrows** with its data name.

Steps to draw Data Flow Diagram

1. **Draw context diagram-**

   It only contains one process node ("**Process 0**") that **generalizes the function of the entire system** in relationship to external entities.

2. **Draw different DFD layers (levels) -**

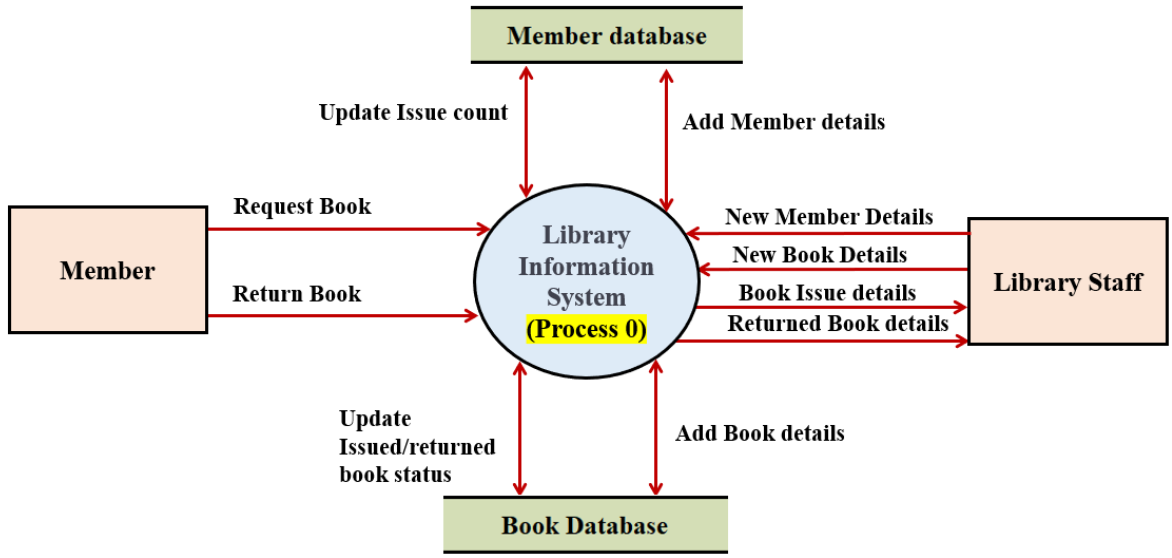
   Context diagram is **followed by various levels** of data flow diagrams.

   A single process node on a high level diagram **can be expanded** to show a more detailed data flow diagram.

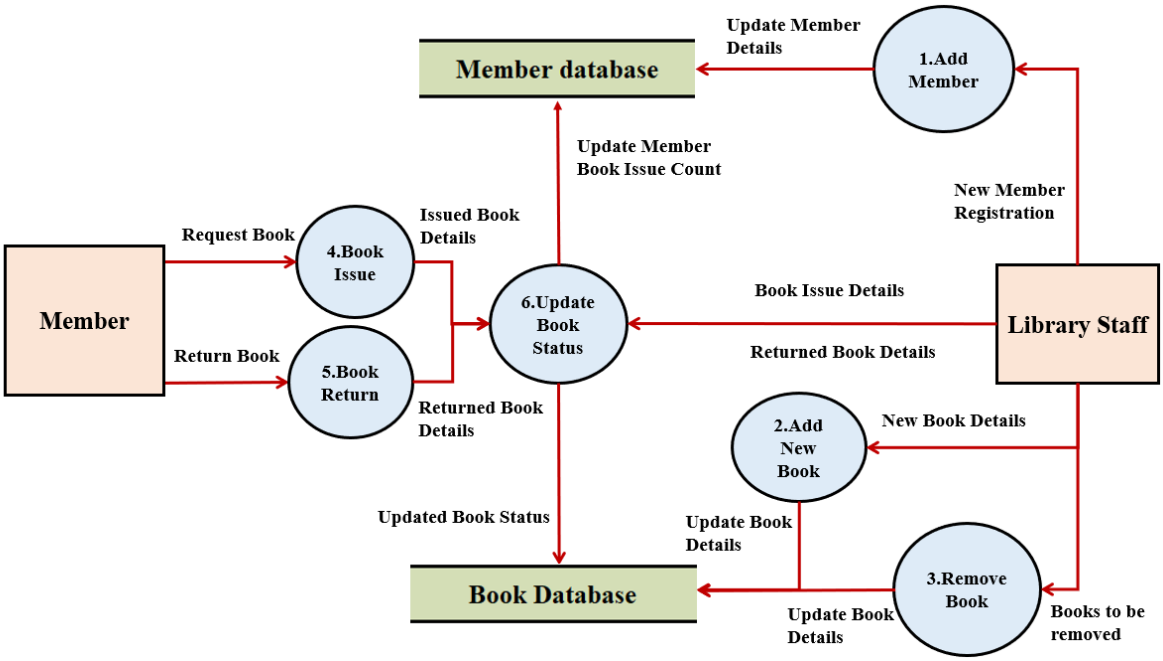
   Each of these processes can be broken into **further processes** until you reach pseudo code.

Example- DFD for Library Information System

Level-0 DFD (context diagram)



Level-1 DFD



http://vlabs.iitkgp.ac.in/se/8/case_study/